

MAT 523: Functions of a Real Variable I

Homework VI

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(1.)

Statement

If E has finite outer measure, then for any $\varepsilon > 0$, E is the disjoint union of a finite number of subsets, each of which has outer measure at most ε .

Proof

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(2.)

Statement

Let E_1, E_2 be measurable sets. Then

$$m(E_1 \cup E_2) + m(E_1 \cap E_2) = m(E_1) + m(E_2).$$

Proof

By the definition of a measurable set, we have that

$$m(E_1) = m(E_1 \cap E_2) + m(E_1 \cap E_2^C)$$

and

$$m(E_2) = m(E_2 \cap E_1) + m(E_2 \cap E_1^C).$$

Adding these equations together yields

$$m(E_1) + m(E_2) = m(E_1 \cap E_2) + m(E_2 \cap E_1) + m(E_1 \cap E_2^C) + m(E_2 \cap E_1^C). \quad (1)$$

Now, we can write $E_1 \cup E_2$ in the following way:

$$\begin{aligned} E_1 \cup E_2 &= (E_1 \setminus E_2) \cup (E_2 \setminus E_1) \cup (E_1 \cap E_2) \\ &= (E_1 \cap E_2^C) \cup (E_2 \cap E_1^C) \cup (E_1 \cap E_2). \end{aligned}$$

Since this is a disjoint union, it follows that

$$m(E_1 \cup E_2) = m(E_1 \cap E_2^C) + m(E_2 \cap E_1^C) + m(E_1 \cap E_2).$$

But then equation (1) becomes

$$m(E_1) + m(E_2) = m(E_1 \cap E_2) + m(E_1 \cup E_2),$$

as required. ■